MIDTERM TASK 3

Normalization Using Power Query

NOTE: you will need to place the needed files/folder in the correct directory if you want to open the power query part (may not even work due to how power query works since the directory needs to be exactly the same)

Step 1: take the raw csv file that needs to be normalized

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	order_id	date	product_id	product_name	price	currency	category_id	category_name	subcategory_id	subcategory_name	store_id	store_name	city_id	city_name	country_id	country
2	CMT-36520	01/02/2020	P002	EconoTread	699.99	GBP	C01	Cardio Equipment	SC01	Treadmills	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
3	CMT-36520	01/02/2020	P020	Door Anchor Straps	24.99	GBP	C03	Functional Fitness	SC10	Suspension Trainers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
4	CMT-36520	01/02/2020	P012	Olympic Flat Bench	179.99	GBP	C02	Strength Training	SC06	Weight Benches	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
5	CMT-36520	01/02/2020	P012	Olympic Flat Bench	179.99	GBP	C02	Strength Training	SC06	Weight Benches	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
6	CMT-36520	01/02/2020	P005	Glide Elite Elliptical	999.99	GBP	C01	Cardio Equipment	SC03	Ellipticals	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
7	MOF-12749	24/07/2020	P004	Recumbent Cycler	599.99	USD	C01	Cardio Equipment	SC02	Exercise Bikes	S002	Muscle Mart	LA	Los Angeles	US	United States
8	MOF-12749	24/07/2020	P003	SpinMaster Bike	799.99	USD	C01	Cardio Equipment	SC02	Exercise Bikes	S002	Muscle Mart	LA	Los Angeles	US	United States
9	MOF-12749	24/07/2020	P016	Heavy Duty Resistanc	39.99	USD	C03	Functional Fitness	SC08	Resistance Bands	S002	Muscle Mart	LA	Los Angeles	US	United States
10	VKL-97451	19/12/2022	P024	Mini Percussion Mas	79.99	GBP	C04	Recovery and Wellne	SC12	Massage Guns	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
11	GJO-36640	14/04/2022	P019	Pro Suspension Trai	129.99	GBP	C03	Functional Fitness	SC10	Suspension Trainers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
12	GJO-36640	14/04/2022	P005	Glide Elite Elliptical	999.99	GBP	C01	Cardio Equipment	SC03	Ellipticals	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
13	GJO-36640	14/04/2022	P017	Cast Iron Kettlebell 1	59.99	GBP	C03	Functional Fitness	SC09	Kettlebells	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
14	GJO-36640	14/04/2022	P026	Extra Thick Exercise	49.99	GBP	C04	Recovery and Wellne	SC13	Yoga Mats	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
15	QNN-19269	02/10/2022	P021	Recovery Roller Deli	39.99	USD	C04	Recovery and Wellne	SC11	Foam Rollers	S001	FitGear Hub	NY	New York	US	United States
16	QNN-19269	02/10/2022	P023	PulseRelief Massage	199.99	USD	C04	Recovery and Wellne	SC12	Massage Guns	S001	FitGear Hub	NY	New York	US	United States
17	QNN-19269	02/10/2022	P020	Door Anchor Straps	24.99	USD	C03	Functional Fitness	SC10	Suspension Trainers	S001	FitGear Hub	NY	New York	US	United States
18	QNN-19269	02/10/2022	P011	Adjustable Bench Pr	249.99	USD	C02	Strength Training	SC06	Weight Benches	S001	FitGear Hub	NY	New York	US	United States
19	TFA-40365	25/10/2021	P026	Extra Thick Exercise	49.99	GBP	C04	Recovery and Wellne	SC13	Yoga Mats	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
20	TFA-40365	25/10/2021	P022	Trigger Point Grid Ro	59.99	GBP	C04	Recovery and Wellne	SC11	Foam Rollers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
21	TFA-40365	25/10/2021	P024	Mini Percussion Mas	79.99	GBP	C04	Recovery and Wellne	SC12	Massage Guns	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
22	TFA-40365	25/10/2021	P011	Adjustable Bench Pr	249.99	GBP	C02	Strength Training	SC06	Weight Benches	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
23	AQN-95165	16/01/2021	P015	FlexForce Band Kit	49.99	USD	C03	Functional Fitness	SC08	Resistance Bands	S003	Strength Central	CH	Chicago	US	United States
24	PSG-64239	16/06/2020	P001	Pro Runner 3000	1299.99	GBP	C01	Cardio Equipment	SC01	Treadmills	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
25	PSG-64239	16/06/2020	P020	Door Anchor Straps	24.99	GBP	C03	Functional Fitness	SC10	Suspension Trainers	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
26	PSG-64239	16/06/2020	P016	Heavy Duty Resistanc	39.99	GBP	C03	Functional Fitness	SC08	Resistance Bands	S005	Active Life Outfitters	MA	Manchester	UK	United Kingdom
27	OAT-63948	25/07/2023	P018	Adjustable Kettlebell	149.99	USD	C03	Functional Fitness	SC09	Kettlebells	S003	Strength Central	CH	Chicago	US	United States
28	OAT-63948	25/07/2023	P025	Premium Yoga Mat	69.99	USD	C04	Recovery and Wellne	SC13	Yoga Mats	S003	Strength Central	CH	Chicago	US	United States
29	IAY-77502	05/02/2020	P026	Extra Thick Exercise	49.99	GBP	C04	Recovery and Wellne	SC13	Yoga Mats	S004	Elite Gym Supply	LO	London	UK	United Kingdom
30	IAY-77502	05/02/2020	P005	Glide Elite Elliptical	999.99	GBP	C01	Cardio Equipment	SC03	Ellipticals	S004	Elite Gym Supply	LO	London	UK	United Kingdom
31	IAY-77502	05/02/2020	P024	Mini Percussion Mas	79.99	GBP	C04	Recovery and Wellne	SC12	Massage Guns	S004	Elite Gym Supply	LO	London	UK	United Kingdom
32	IAY-77502	05/02/2020	P003	SpinMaster Bike	799.99	GBP	C01	Cardio Equipment	SC02	Exercise Bikes	S004	Elite Gym Supply	LO	London	UK	United Kingdom
33	IAY-77502	05/02/2020	P014	Leg Press Machine	899.99	GBP	C02	Strength Training	SC07	Weight Machines	S004	Elite Gym Supply	LO	London	UK	United Kingdom
34	JOH-53793	02/01/2023	P013	Multi-Gym Station	1499.99	GBP	C02	Strength Training	SC07	Weight Machines	S004	Elite Gym Supply	LO	London	UK	United Kingdom

Step 2: proceed with cleaning/normalization

	order_id	date	product_id	product_name	price	currency	category_id	category_name
1	ALM-27496	06/06/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
2	LTW-72095	11/09/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
3	RXK-83093	01/03/2022	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
4	HEZ-96940	03/05/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
5	IAT-97463	15/02/2022	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
6	ASU-35409	05/08/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
7	KCA-65476	29/07/2022	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
8	UDR-01293	15/12/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
9	APD-39715	03/11/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
10	AEI-61734	05/04/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
11	MXG-29539	20/05/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
12	PMS-09120	12/03/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
13	UIF-43588	10/06/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
14	VNR-35178	09/06/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
15	NZR-34932	17/06/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
16	TFA-40365	25/10/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
17	AJK-07698	17/04/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
18	EKK-86559	03/12/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
19	DML-37731	24/03/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
20	THO-06946	17/10/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
21	TYY-21104	21/03/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
22	CNN-38516	02/01/2022	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
23	PKA-92985	16/05/2023	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
24	KQC-58283	27/05/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
25	STE-32865	18/07/2022	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
26	PHQ-11585	08/12/2021	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
27	HII-26166	19/11/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training
28	HIS-46378	16/02/2020	P011	Adjustable Bench Pro	249.99	GBP	C02	Strength Training

Step 3: Create the needed dimension tables

(the excel file also has all the fact and dimension tables in actual sheets)

	product_id	product_name	price	subcategory_id
1	P001	Pro Runner 3000	1299.99	SC01
2	P002	EconoTread	699.99	SC01
3	P003	SpinMaster Bike	799.99	SC02
4	P004	Recumbent Cycler	599.99	SC02
5	P005	Glide Elite Elliptical	999.99	SC03
6	P006	Compact Cross Trainer	549.99	SC03
7	P007	AquaRow 500	899.99	SC04
8	P008	AirRow Pro	1099.99	SC04
9	P009	PowerLift Dumbbell Set	299.99	SC05
10	P010	Cast Iron Plate Set	199.99	SC05
11	P011	Adjustable Bench Pro	249.99	SC06
12	P012	Olympic Flat Bench	179.99	SC06
13	P013	Multi-Gym Station	1499.99	SC07
14	P014	Leg Press Machine	899.99	SC07
15	P015	FlexForce Band Kit	49.99	SC08
16	P016	Heavy Duty Resistance Tubes	39.99	SC08
17	P017	Cast Iron Kettlebell 16kg	59.99	SC09
18	P018	Adjustable Kettlebell	149.99	SC09
19	P019	Pro Suspension Trainer	129.99	SC10
20	P020	Door Anchor Straps	24.99	SC10
21	P021	Recovery Roller Deluxe	39.99	SC11
22	P022	Trigger Point Grid Roller	59.99	SC11
23	P023	PulseRelief Massage Gun	199.99	SC12
24	P024	Mini Percussion Massager	79.99	SC12
25	P025	Premium Yoga Mat	69.99	SC13
26	P026	Extra Thick Exercise Mat	49.99	SC13

	category_id	category_name
1	C01	Cardio Equipment
2	C03	Functional Fitness
3	C02	Strength Training
4	C04	Recovery and Wellness

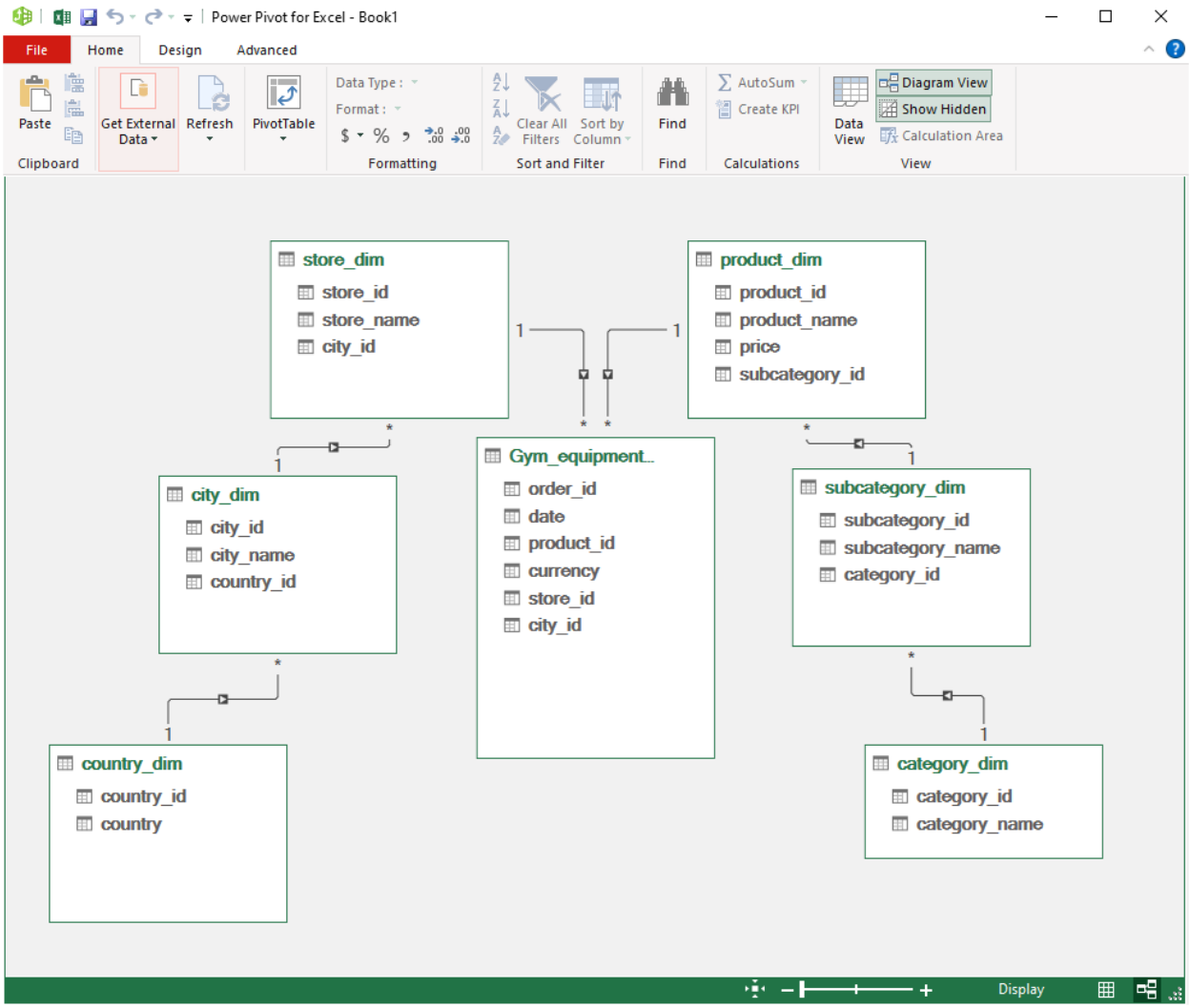
	subcategory_id	subcategory_name	category_id
1	SC01	Treadmills	C01
2	SC10	Suspension Trainers	C03
3	SC06	Weight Benches	C02
4	SC03	Ellipticals	C01
5	SC02	Exercise Bikes	C01
6	SC08	Resistance Bands	C03
7	SC12	Massage Guns	C04
8	SC09	Kettlebells	C03
9	SC13	Yoga Mats	C04
10	SC11	Foam Rollers	C04
11	SC07	Weight Machines	C02
12	SC04	Rowing Machines	C01
13	SC05	Free Weights	C02

	store_id	store_name	city_id
1	S001	FitGear Hub	NY
2	S002	Muscle Mart	LA
3	S003	Strength Central	CH
4	S004	Elite Gym Supply	LO
5	S005	Active Life Outfitters	MA

	city_id	city_name	country_id
1	CH	Chicago	US
2	LA	Los Angeles	US
3	LO	London	UK
4	MA	Manchester	UK
5	NY	New York	US

	country_id	country
1	UK	United Kingdom
2	US	United States

Step 4: create the fact table

	order_id	date	product_id	currency	store_id	city_id
1	CMT-36520	01/02/2020	P002	GBP	S005	MA
2	CMT-36520	01/02/2020	P020	GBP	S005	MA
3	CMT-36520	01/02/2020	P012	GBP	S005	MA
4	CMT-36520	01/02/2020	P012	GBP	S005	MA
5	CMT-36520	01/02/2020	P005	GBP	S005	MA
6	MOF-12749	24/07/2020	P004	USD	S002	LA
7	MOF-12749	24/07/2020	P003	USD	S002	LA
8	MOF-12749	24/07/2020	P016	USD	S002	LA
9	VKL-97451	19/12/2022	P024	GBP	S005	MA
10	GJO-36640	14/04/2022	P019	GBP	S005	MA
11	GJO-36640	14/04/2022	P005	GBP	S005	MA
12	GJO-36640	14/04/2022	P017	GBP	S005	MA
13	GJO-36640	14/04/2022	P026	GBP	S005	MA
14	QNN-19269	02/10/2022	P021	USD	S001	NY
15	QNN-19269	02/10/2022	P023	USD	S001	NY
16	QNN-19269	02/10/2022	P020	USD	S001	NY
17	QNN-19269	02/10/2022	P011	USD	S001	NY
18	TFA-40365	25/10/2021	P026	GBP	S005	MA
19	TFA-40365	25/10/2021	P022	GBP	S005	MA
20	TFA-40365	25/10/2021	P024	GBP	S005	MA
21	TFA-40365	25/10/2021	P011	GBP	S005	MA
22	AQN-95165	16/01/2021	P015	USD	S003	CH
23	PSG-64239	16/06/2020	P001	GBP	S005	MA
24	PSG-64239	16/06/2020	P020	GBP	S005	MA
25	PSG-64239	16/06/2020	P016	GBP	S005	MA
26	OAT-63948	25/07/2023	P018	USD	S003	CH
27	OAT-63948	25/07/2023	P025	USD	S003	CH
28	IAY-77502	05/02/2020	P026	GBP	S004	LO
29	IAY-77502	05/02/2020	P005	GBP	S004	LO

Step 5: create an ERD data model